CLASSROOM PLANETARY CLOCK & INTERVAL SYSTEM

Kinesthetic Time, Epicyclic 12:1 Gears & Modular Arithmetic

SERIES 1 • CARD 03 / 12 • DOMAIN: KINEMATIC GEAR RATIOS & MODULAR ARITHMETIC

CARD 03 / 12 • TIME-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

EPICYCLIC 12:1 GEARS & MODULO ARITHMETIC

Time is continuous rotational motion & modular math.

StackCalc Clock features an epicyclic gearset:

12:1 Exact Planetary Reduction Ratio

When minute hand turns 360°, hour hand advances 30°!

CALIBRATED 60-MINUTE / 12-HOUR DATUM

- Minute Unit Datum: 1 Minute = 6° dial sector
- Hour Mark: 1 Hour Mark = 30° arc (12 total sectors)
- Quadrant: 1/4 Hour Wedge = 90° right-angle sector

PLANETARY GEARS: 12:1 RATIO AT 3:30 (105°)



12:1 EPICYCLIC GEARS

360° min → 30° hour

At 3:30: Hour hand = 105°

15-Min Quadrant Wedge:

90° arc = 1/4 of an hour

Modulo 12 & Modulo 60

2. KIT PARTS INVENTORY (18 PIECES)

COMPONENT	QTY	SPEC	COLOR
Clock Frame	1x	105.0 mm Dial	Slate Gray
12:1 Gears	1x	Planetary	Gold / Blue
15-Min Wedges	4x	90° Quadrant	Sky Blue
30-Min Wedges	2x	180° Sector	Royal Purple
5-Min Wedges	8x	30° Sector	Emerald Green
Stand & Lid	1x	Stand & Lid	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 16 Clock & Wedge Components + Storage Bin + Stand Lid = 18 Parts

FRACTION-ALIGNED TIME PALETTE

- Blue Wedges: 15-minute quadrants (90° right angles).
- Purple Wedges: 5-minute segments (30° hour steps).
- Emerald Face: Calibrated clock dial with 6° ticks.
- High-Visibility Hands: White minute & orange hour hands.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Seat the geared clock frame into its bottom cradle. Drop the 14 colored wedges into their labeled radial lid bays; any missing wedge leaves an instantly empty bay. Snap lid shut.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Clock arithmetic is modulo 12 and modulo 60 governed by strict continuous gear kinematics.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: ELAPSED TIME & PLANETARY GEAR RATIOS

Step-by-Step Hands-On Tutorial: 3:30 + 0:45 = 4:15 & Modular Time

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LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/classroom-clock/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 3.MD.A.1 (Elapsed Time)
- CCSS 4.MD.A.2 (Time Intervals)
- TEKS 111.5.b.7.C (Elapsed Time...)

Grades 2–6 Elapsed Time & Modular Math

GUIDING INQUIRY

How far does the hour hand travel when the minute hand turns one full circle around the dial?

1 STEP 1: SET START TIME TO 3:30

Rotate the minute hand to 6 (bottom of dial, 30 minutes).

Observe the internal gears: the hour hand advances smoothly halfway between 3 and 4 (exactly 105° on the dial).

2 STEP 2: INSERT 15-MINUTE WEDGES

We need to add 45 minutes of elapsed time.

Insert three 15-minute wedges (3 × 15 min = 45 min).

Wedges cover 3 × 90° = 270° angular arc around dial.

Notice: The wedges bridge past the 12

3-4 STEPS 3 & 4: ADVANCE GEARS & READ RESULT

3. ADVANCE MINUTE HAND: Turn hand clockwise across the 3 wedges past 4:00 to 4:15!

4. READ FINAL TIME: Hour hand at 4:15; minute hand at 3 (15 min).

Calculator Verification: Press Clock HMS Icon ([🕒]/[g][▸HMS]/[f][▸HR]) and run 3.30 [f] [▸HR] 0.45 [f] [▸HR] [+] [g] [▸HMS] → 4.15.

5 CORE PRINCIPLE: TIME IS CONTINUOUS KINEMATIC RATIO

Digital clocks make students think hour hands jump instantly when minutes reach 60. The planetary gears physically prove that time is a continuous ratio! Because 45 minutes is 3/4 of an hour, the hour hand moves through exactly 3/4 of its 30° sector (22.5°), landing precisely at

★ EXTENSION INVESTIGATION: ELAPSED TIME: 8:45 + 1:30

Set clock to 8:45. Add two 45-minute wedges (90 min). Where does the planetary gear place the hour hand? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

STACKCALC ARCHIVAL INSTRUMENT LABS

Interactive Mathematics Manipulative Ecosystem • Mountain View, CA • BPA-Free

PARTS & REFILL PORTAL

stackcalc.io/parts

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH

Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC

Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY

Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING

Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5

Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC

[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS

Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS

Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING

Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1

Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION

Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS

Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED

Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS

Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4

Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:

Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:

Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR STORAGE DETENTS:

Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:

Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

TIME-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:

All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:

Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] ZERO-RATTLE DETENT LOCK:

Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:

2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:

100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:

Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 18 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io

stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California